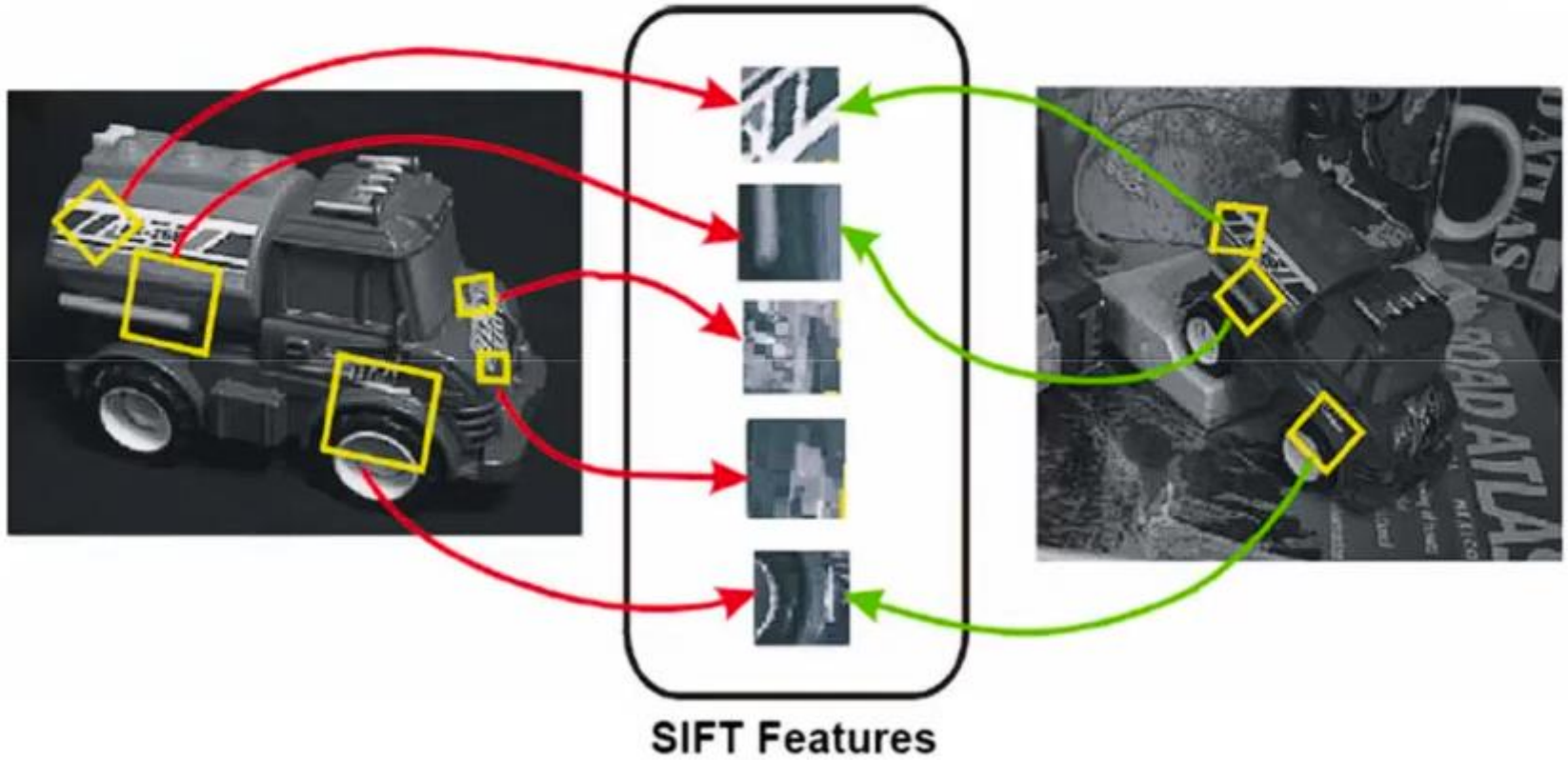
SIFT

SIFT: Scale Invariant Feature Transform

- David G. Lowe, *Distinctive Image Features from Scale-invariant Keypoints*, IJCV 2004
  - Over 50000 citations
- Transforms image data into scale-invariant coordinates
- Fundamental to many core vision problems/applications:
  - Recognition, Motion tracking, Multiview geometry

SIFT: Invariant Local Features

Image content is transformed into local feature coordinates that are invariant to translation, rotation, scale, shear.



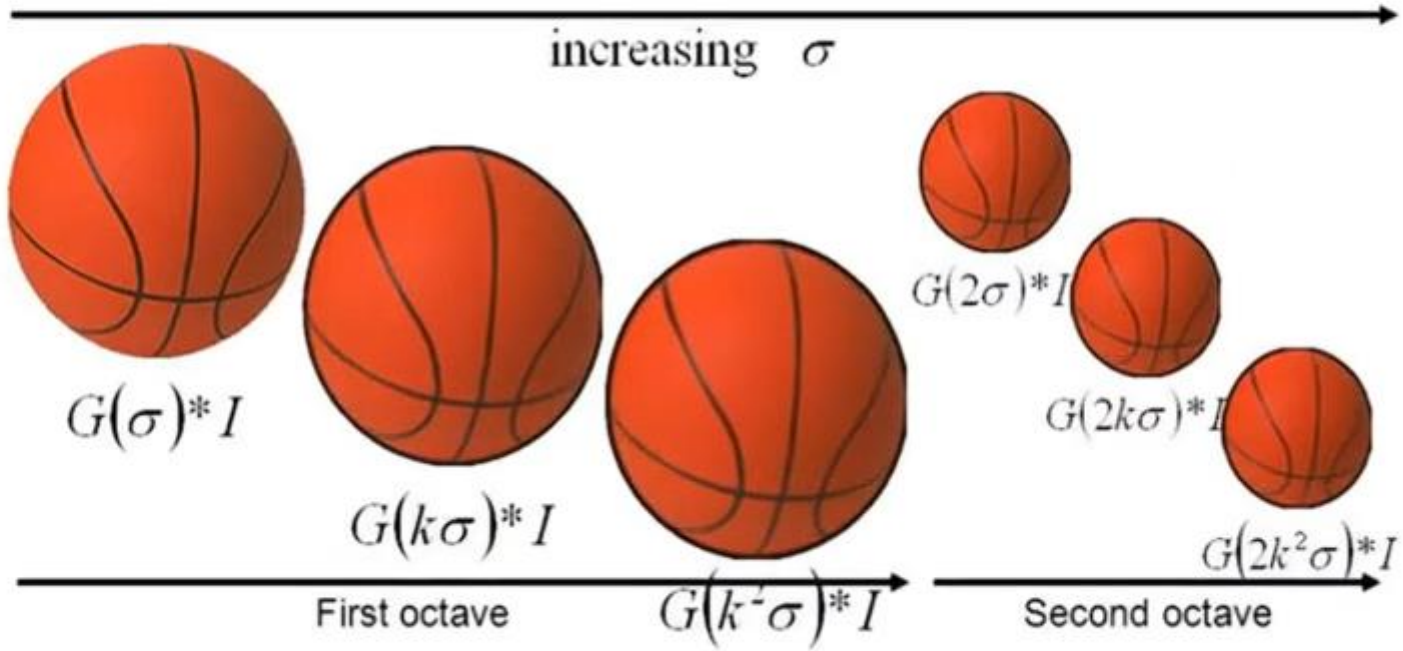
Credit: Mubarak Shah, University of Central Florida

SIFT

- **Step 1: Scale-space Extrema Detection** - Detect interesting points (invariant to scale and orientation) using DOG.
- **Step 2: Keypoint Localization** - Determine location and scale at each candidate location, and select them based on stability.
- **Step 3: Orientation Estimation** - Use local image gradients to assign orientation to each localized keypoint. Preserve orientation, scale and location for each feature.
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SIFT: Scale-space Extrema Detection

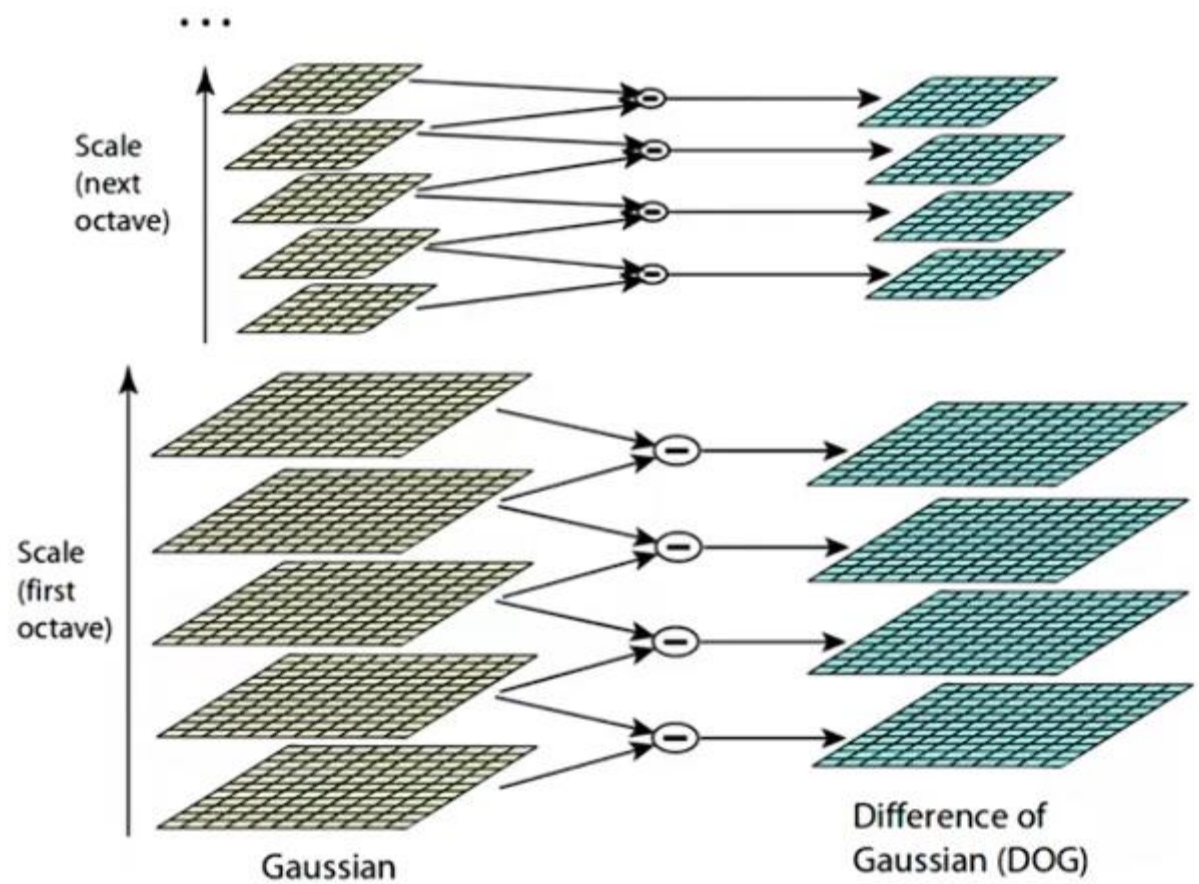
Constructing Scale Space



Credit: Ofir Pele

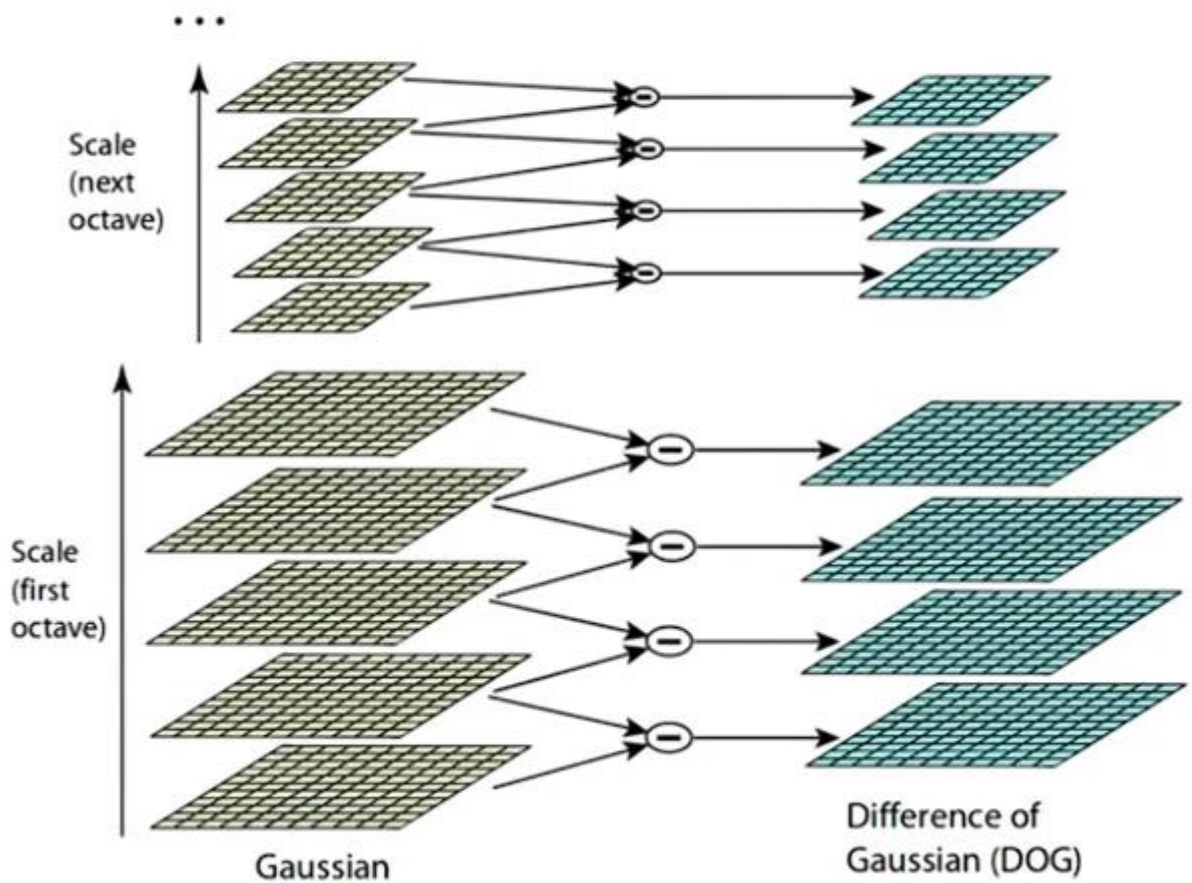
SIFT: Scale-space Extrema Detection

Difference of Gaussians



SIFT: Scale-space Extrema Detection

Difference of Gaussians

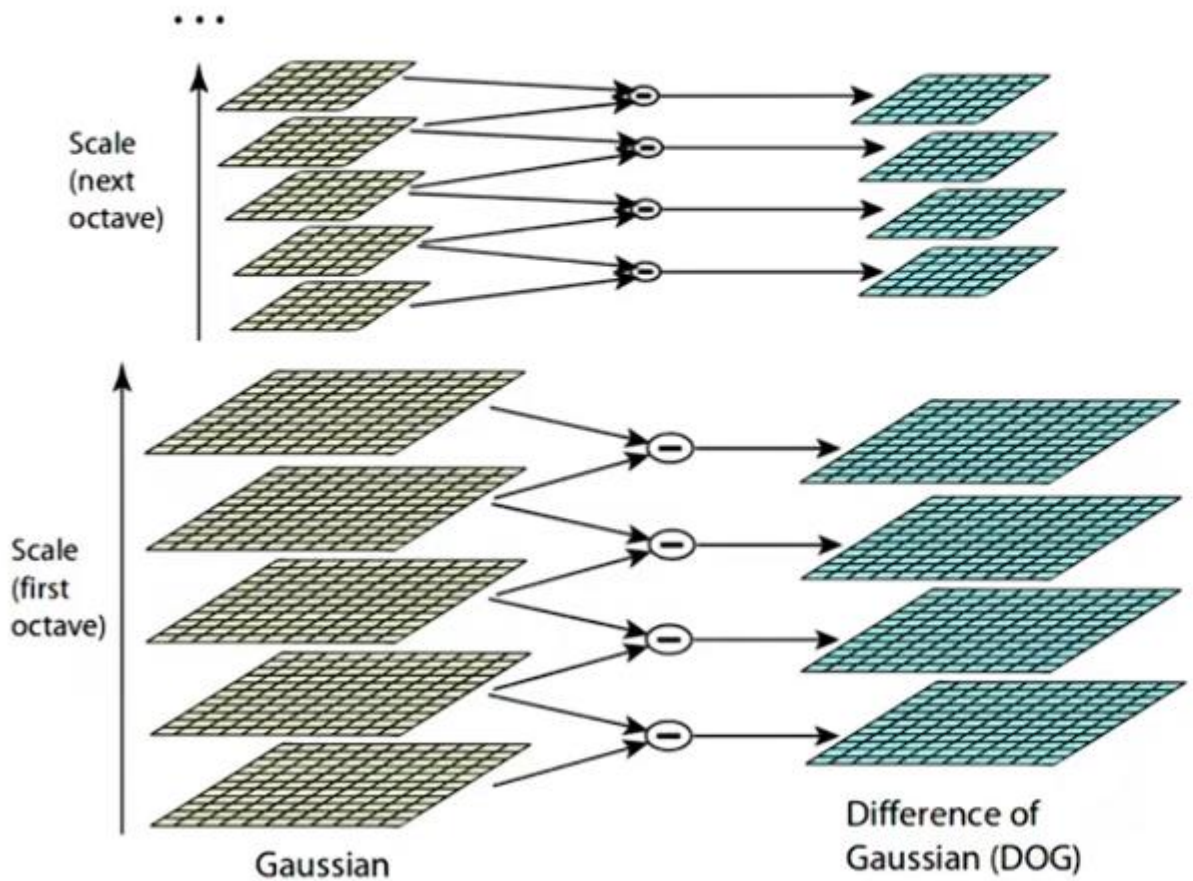


$$D(x, y, \sigma) = \hat{I}(x, y, \sigma) - \hat{I}(x, y, k\sigma)$$

where $\hat{I}(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$

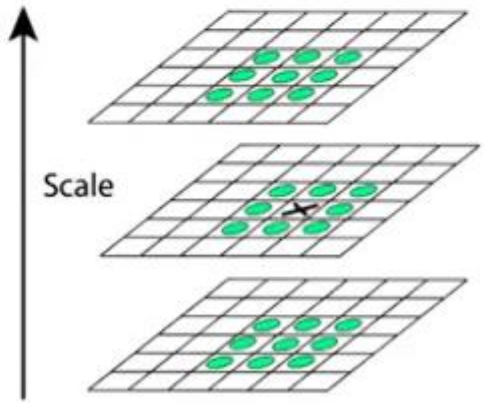
SIFT: Scale-space Extrema Detection

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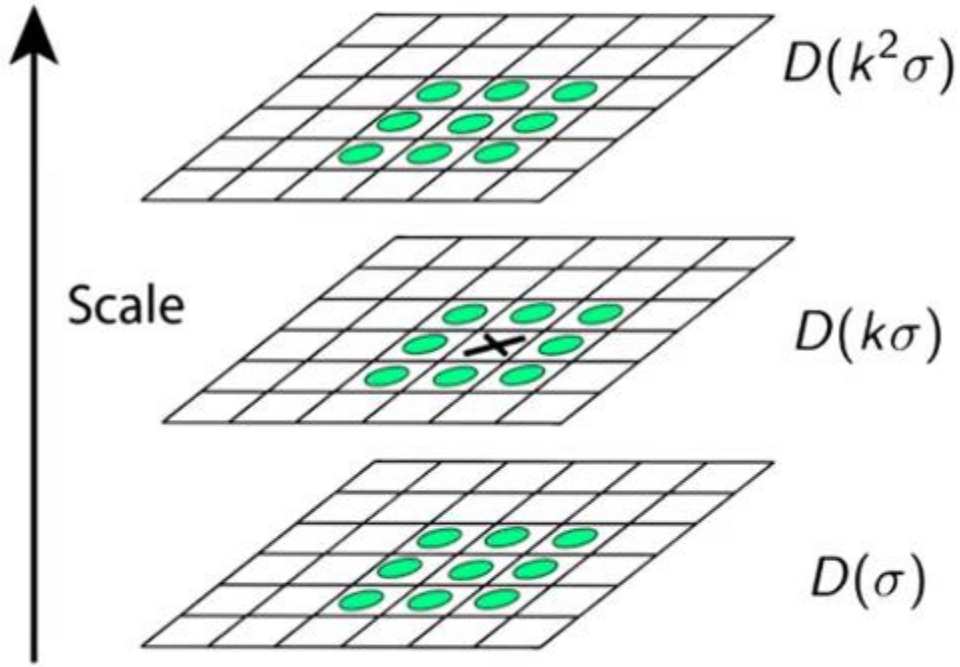
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Credit: "Distinctive Image Features from Scale-Invariant Points", IJCV 2004

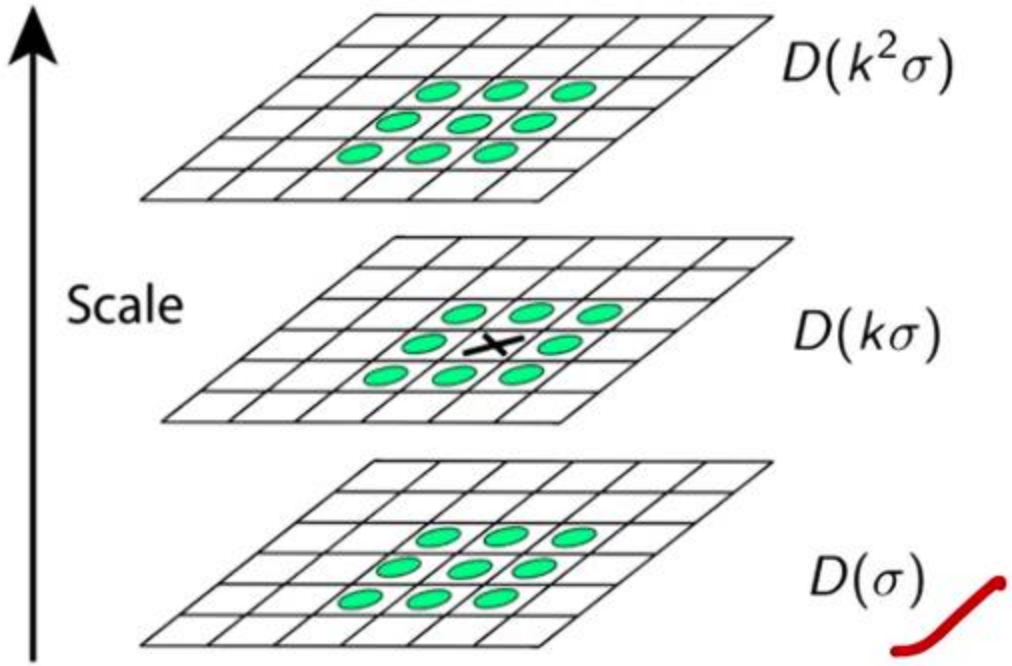
SIFT: Scale-space Extrema Detection



- Compare a pixel (X) with 26 pixels in current and adjacent scales (Green Circles)
- Select a pixel (X) if it is larger/smaller than all 26 pixels

Credit: "Distinctive Image Features from Scale-Invariant Points", IJCV 2004

SIFT: Scale-space Extrema Detection



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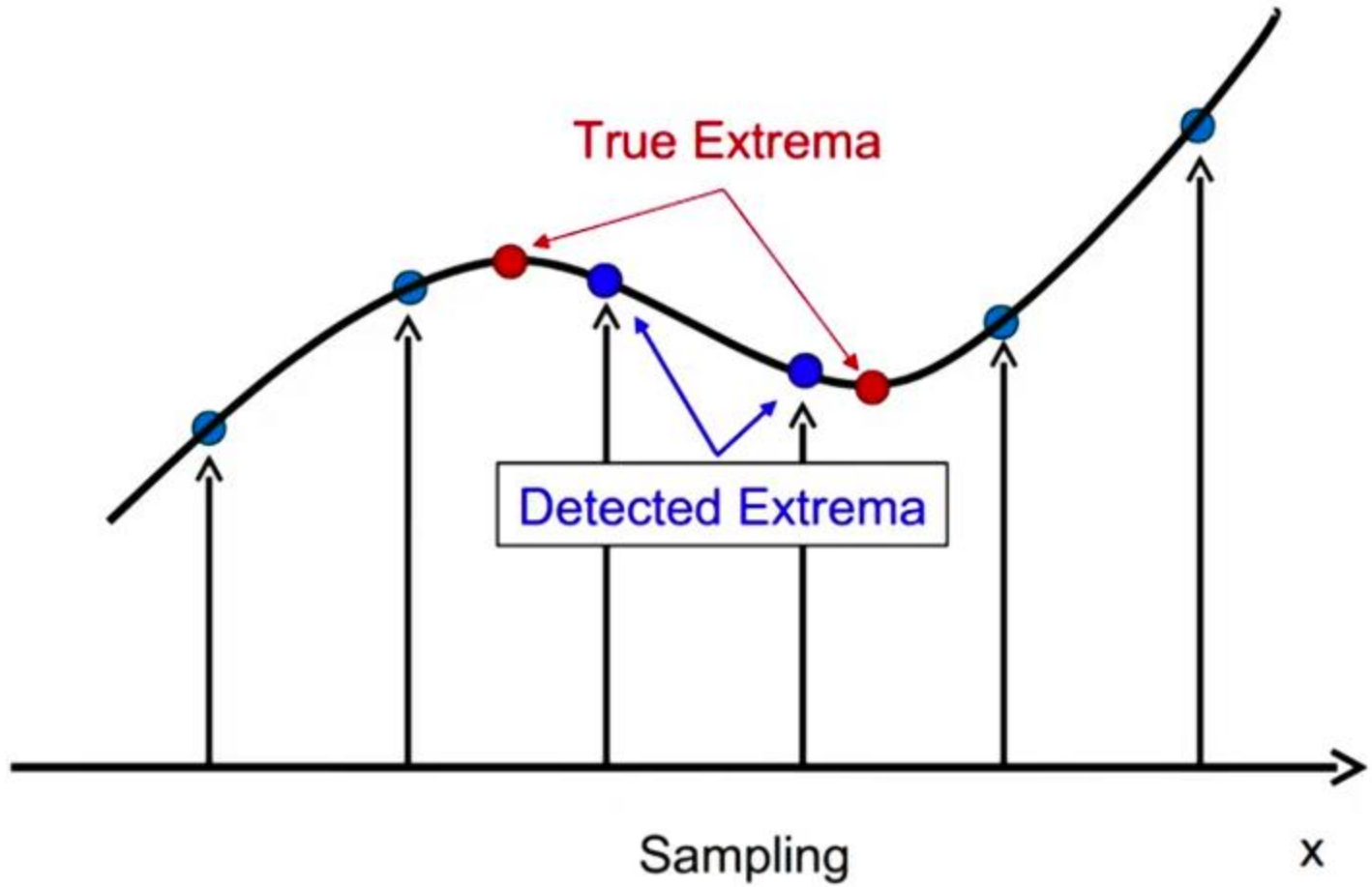
Credit: "Distinctive Image Features from Scale-Invariant Points", IJCV 2004

SIFT Algorithm Stages

- **Step 1: Scale-space extrema Detection** - Detect interesting points (invariant to scale and orientation) using DOG.
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SIFT: Keypoint Localization

The Problem:



Credit: Ofir Pele

SIFT: Keypoint Localization

The Solution:

- Use Taylor series expansion of the scale-space function:

$$D(\mathbf{s}_0 + \Delta\mathbf{s}) = D(\mathbf{s}_0) + \frac{\partial D}{\partial \mathbf{s}}^T \Big|_{\mathbf{s}_0} \Delta\mathbf{s} + \frac{1}{2}\Delta\mathbf{s}^T \frac{\partial^2 D}{\partial \mathbf{s}^2}\Big|_{\mathbf{s}_0} \Delta\mathbf{s}$$

where $\mathbf{s}_0 = (x_0, y_0, \sigma_0)^T$ and $\Delta\mathbf{s} = (\delta x, \delta y, \delta\sigma)^T$

SIFT: Keypoint Localization

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- The location of the extremum, $\hat{\mathbf{s}}$, is determined by taking the derivative of this function with respect to $\mathbf{s}$ and setting it to zero:

$$\hat{\mathbf{s}} = -\left(\frac{\partial^2 D}{\partial \mathbf{s}^2}\Big|_{\mathbf{s}_0}\right)^{-1} \frac{\partial D}{\partial \mathbf{s}}\Big|_{\mathbf{s}_0}$$

- Next, reject low contrast points and points that lie on the edge

SIFT: Keypoint Localization

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- Next, reject low contrast points and points that lie on the edge
- Low contrast points elimination:
  - Reject keypoint if $D(\hat{\mathbf{s}})$ is smaller than 0.03 (assuming image values have been normalized to [0,1])

SIFT: Keypoint Localization

- Reject points with strong edge response in one direction only
- Edge Elimination - How?

SIFT: Keypoint Localization

- Reject points with strong edge response in one direction only
- Edge Elimination - How?
  - Similar to Harris corner detector!
  - SIFT instead uses Hessian
- Compute Hessian of D (principal curvature)

$$H = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix}$$

α : largest eigenvalue(λ_{max})

β : smallest eigenvalue(λ_{min})

$$Tr(H) = D_{xx} + D_{yy} = \alpha + \beta$$

$$Det(H) = D_{xx}D_{yy} - D_{xy}^2 = \alpha\beta$$

SIFT: Keypoint Localization

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- Evaluate ratio

$$\frac{Tr(H)^2}{Det(H)} = \frac{(\alpha+\beta)^2}{\alpha\beta} = \frac{(r\beta+\beta)^2}{r\beta^2}$$

$$\frac{Tr(H)^2}{Det(H)} = \frac{(r+1)^2}{r} \text{ where } r = \frac{\alpha}{\beta}$$

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- This quantity is minimum when r = 1 (eigenvalues are equal)
- Reject keypoint if: $\frac{Tr(H)^2}{Det(H)} >$ a threshold (Original SIFT uses r = 10)

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SIFT: Orientation Estimation

- Why? To achieve rotation invariance

SIFT: Orientation Estimation

- Why? To achieve rotation invariance
- Use scale of point to choose correct image:

$$\hat{I}(x, y) = G(x, y, \sigma) * I(x, y)$$

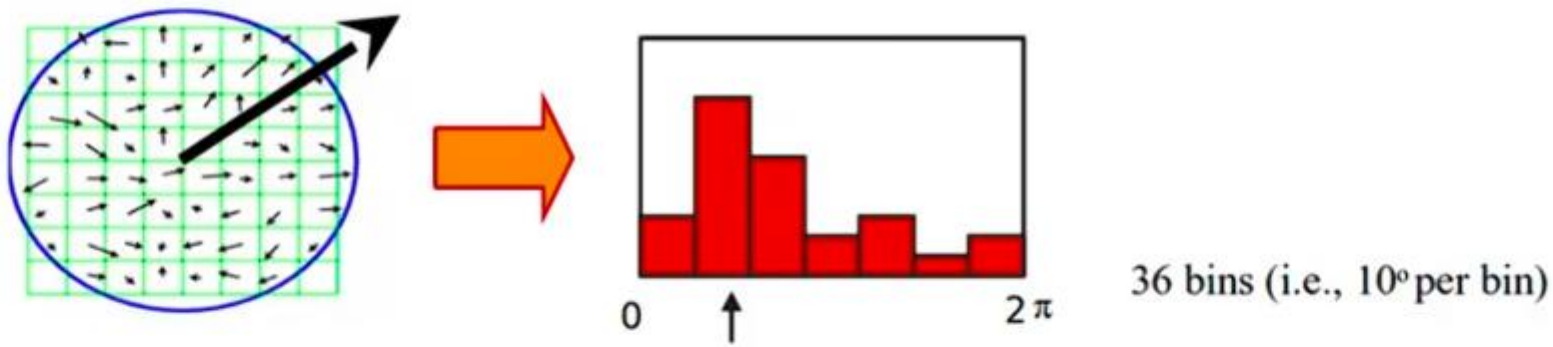
- Compute gradient magnitude and orientation using finite differences:

$$m(x, y) = \sqrt{(\hat{I}(x+1, y) - \hat{I}(x-1, y))^2 + (\hat{I}(x, y+1) - \hat{I}(x, y-1))^2}$$

$$\theta(x, y) = \tan^{-1}\left(\frac{(\hat{I}(x, y+1) - \hat{I}(x, y-1))}{(\hat{I}(x+1, y) - \hat{I}(x-1, y))}\right)$$

SIFT: Orientation Estimation

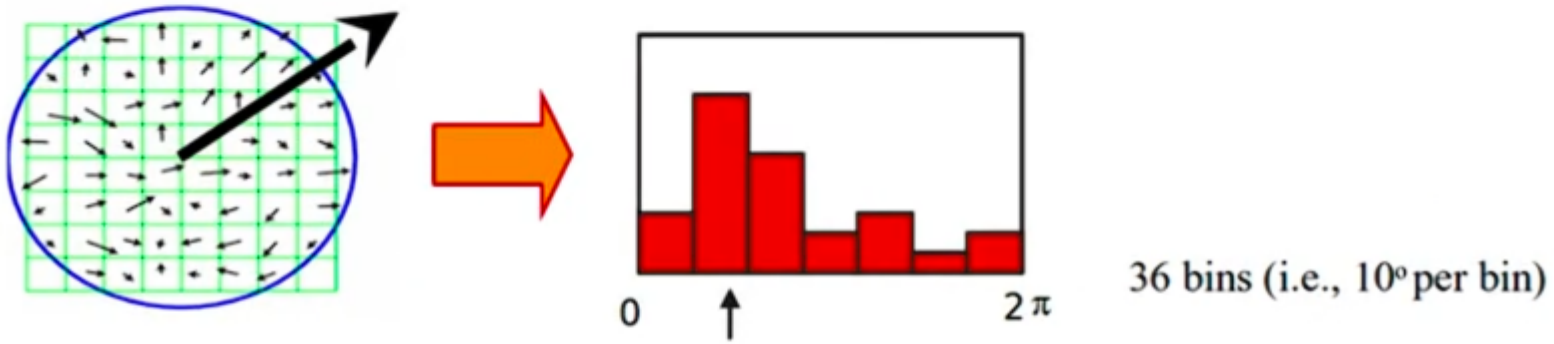
- Create histogram of gradient directions, within a region around the keypoint, at selected scale:



- Histogram entries are weighted by:
  - gradient magnitude, and
  - a Gaussian function with σ equal to 1.5 times scale of the keypoint

SIFT: Orientation Estimation

- Create histogram of gradient directions, within a region around the keypoint, at selected scale:



- Histogram entries are weighted by:
  - gradient magnitude, and
  - a Gaussian function with σ equal to 1.5 times scale of the keypoint
- Select **the peak as direction of keypoint**
- Introduce additional key points at same location if another peak is within 80% of max peak of histogram with different direction

Credit: Svetlana Lazebnik, UIUC

SIFT





From 233x189 original image to 832 DoG Extrema

SIFT





From 832 DoG Extrema to 729 keypoints after low contrast threshold

SIFT





From 729 keypoints to 536 keypoints after testing ratio based on Hessian

SIFT Algorithm Stages

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SIFT: Keypoint Descriptor

- Compute gradient at each pixel in a 16×16 window around the detected keypoint, using the appropriate leve as detected.

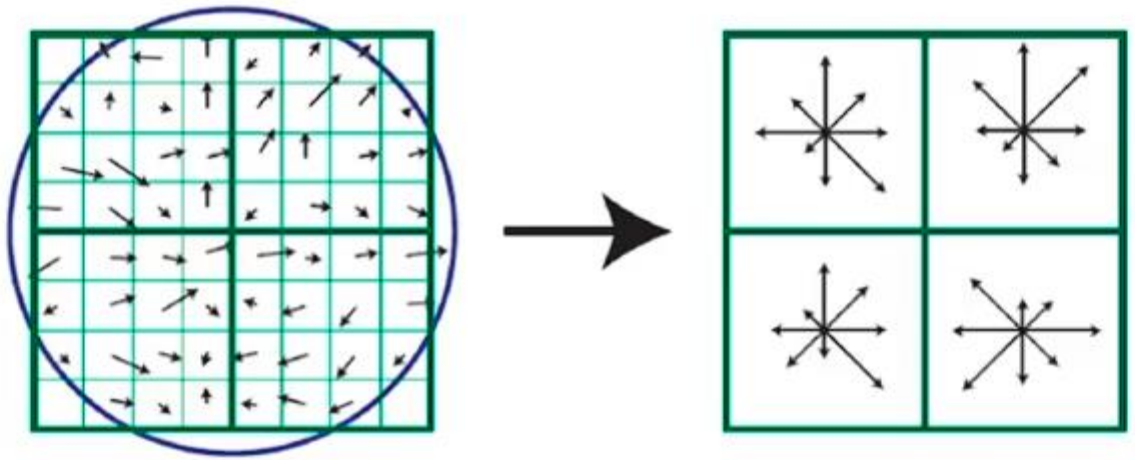


Image gradients

Keypoint descriptor

SIFT: Keypoint Descriptor

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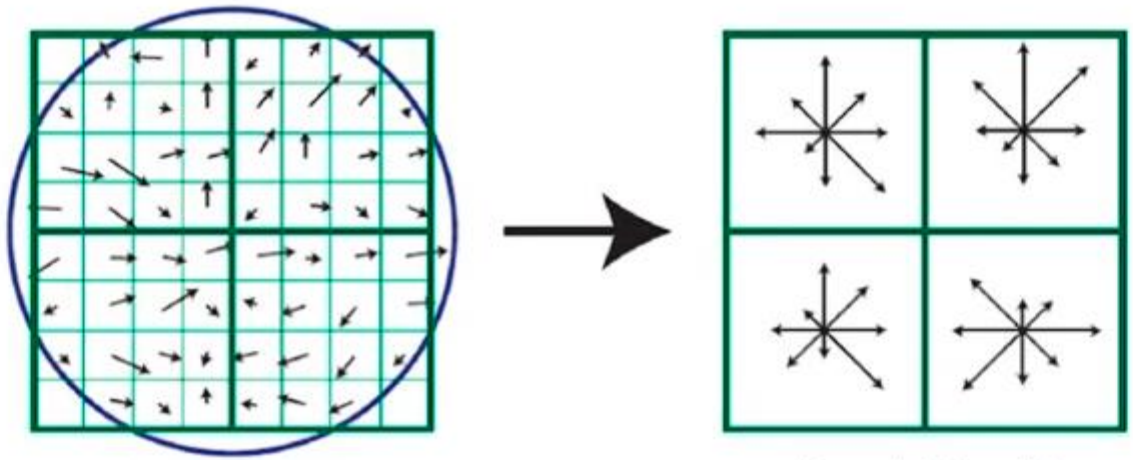


Image gradients

Keypoint descriptor

- Downweight gradients by a Gaussian fall-off function (blue circle) to reduce the influence of gradients far from the center.

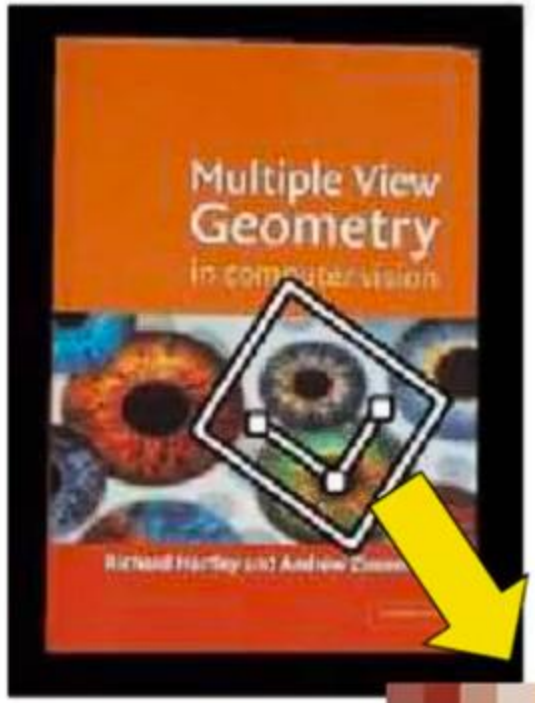
SIFT: Keypoint Descriptor

- The resulting 128 non-negative values form a raw version of the SIFT descriptor vector.
- To reduce the effects of contrast or gain (additive variations are already removed by the gradient), the 128-D vector is normalized to unit length.
- To further make the descriptor robust to other photometric variations, values are clipped to 0.2 and the resulting vector is once again renormalized to unit length.

SIFT

- Extraordinarily robust feature detection
- Changes in viewpoint: up to about 60 degree out of plane rotation
- Changes in illumination: sometimes even day vs night (below)
- Fast and efficient – can run in real-time

SIFT: Invariances(Geometric Transformations)









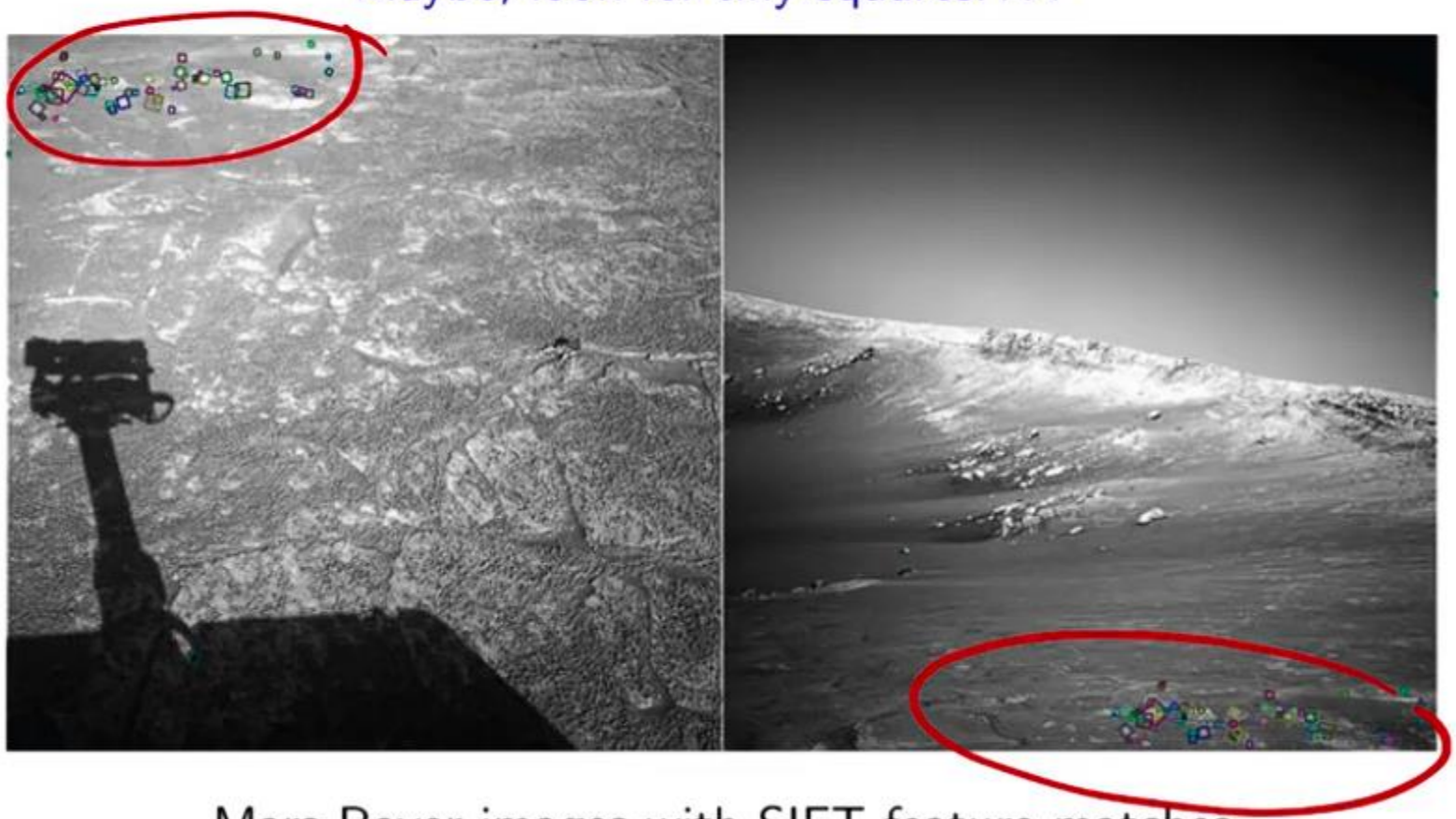
e.g. scale,
translation,
rotation

SIFT: Invariances(Photometric Transformations)



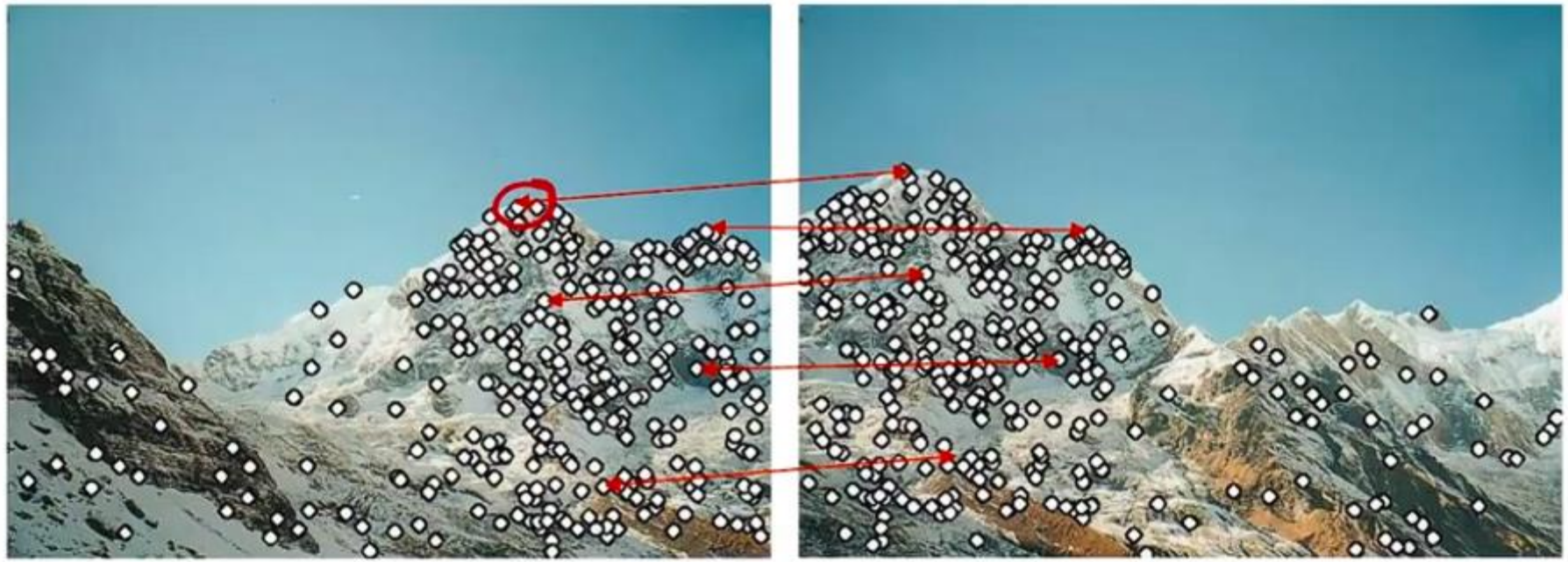
SIFT: Example

Maybe, look for tiny squares. . . ?



Mars Rover images with SIFT feature matches

SIFT Applications: Image Stitching



- Detect feature points in both images.
- Find corresponding pairs of feature points.

SIFT Applications: Image Stitching



- Detect feature points in both images.
- Find corresponding pairs of feature points.
- Use the pairs the align the images.

Homework

Readings

- Section 3.5, Szeliski, *Computer Vision: Algorithms and Applications*
- For more information on SURF:
  - OpenCV-Python Tutorials : Introduction to SURF
  - Wikipedia: Speeded up robust features
- Multi-Scale Oriented Patches
- Other links provided on respective slides

Questions

- Which descriptor performs better? SIFT or MOPS?
- Why is SIFT descriptor better than Harris Corner Detector?

Reference

- https://www.youtube.com/watch?v=gEoWMLXk0Ys&list=PLyqSpQzTE6M_PI-rIz4O1jEgffhJU9GgG&index=11